

LEGAL AND OPEN-SOURCE LICENCES

iTantra: speech models, licences and offline deployment

iTantra is licensed GPL-3.0-or-later.

The app links eSpeak NG, which is GPL-3.0-or-later, through JNI. The combined application is therefore distributed under GPL-3.0-or-later (see LICENSE). Every other component below keeps its own GPL-compatible licence and notices. [7, 12, 13]

1. Offline by design

STT turns speech into text, TTS reads text aloud and VAD detects speech and pauses. All of them run on the phone. There is **no hosted STT, TTS, translation or other cloud API**, and no audio or text is sent to a server. Messages travel only over local Wi-Fi, Wi-Fi Direct or Bluetooth links.

Models are not bundled in the base APK. Speech-recognition and neural-voice packs are imported from local storage (.itpack, SHA-256 checked against models/models.lock.json). The preloaded demo APK bundles English and Hindi STT plus the English and Hindi neural voices.

2. Speech components

COMPONENT	USE / LANGUAGES	SIZE	LICENCE
Moonshine Small Streaming	STT: English default.	142.32 MB	MIT (Section 1 of the pinned publisher LICENSE covers all streaming models). [1]
Moonshine Tiny Streaming	STT: smaller English option for low-end devices.	45.25 MB	MIT, same publisher commit and licence as Small. [1]
AI4Bharat IndicConformer (INT8)	STT: Bengali, Gujarati, Hindi, Kannada, Malayalam, Marathi, Odia*, Tamil, Telugu.	197.66 MB per language	Original AI4Bharat models: MIT. The community ONNX export repository declares Apache-2.0. Both notices are retained. [2-4]
AI4Bharat Indic-TTS FastPitch + HiFi-GAN V1 neural voices	Optional TTS packs for all ten languages. Acoustic: EchoBharat int8 conversion. Vocoder: re-exported here as float32.	118.93 MB per language (Odia 183.59 MB)	MIT as declared by the EchoBharat publisher (licenses/indictts-echobharat-LICENSE.txt). The upstream AI4Bharat/Indic-TTS repository ships no LICENSE file, so the MIT term is not confirmed upstream and must be verified before release. Both conversions are declared modifications. [5, 6]
eSpeak NG 1.53.0	Built-in TTS for all ten languages: default, fallback and emergency voice. Formant synthesis.	3.24 MB total	GPL-3.0-or-later; separately licensed files keep COPYING.APACHE, COPYING.BSD2 and COPYING.UCD. [7]
Silero VAD (INT8)	Speech/pause detection for push-to-talk and hands-free.	213 KB	MIT, Copyright (c) 2020-present Silero Team. [8]

*Odia STT is a local CTC export of the MIT model and remains experimental; the Odia voice is a locally exported split graph. Export-tool licences (AI4Bharat NeMo fork, Apache-2.0) are host-only and not bundled. Accuracy and voice quality are not yet validated.

Sizes use decimal MB/KB and are summed from models/models.lock.json (model files, tokens and configuration). The eSpeak total is the native engine plus uncompressed voice data. None of these is a RAM measurement.

3. Runtime and app libraries

COMPONENT	ROLE	LICENCE
sherpa-onnx 1.13.8 AAR	Runs IndicConformer STT and Silero VAD.	Apache-2.0. [9]
ONNX Runtime 1.28.2	The libonnxruntime.so inside the sherpa-onnx AAR. The neural-voice bridge (libitantra_tts.so, app code) loads this same library; no second runtime is packaged for neural voices .	MIT. [10]
Moonshine Android runtime 0.1.5	Runs Moonshine STT. Repackaged for arm64 with its own isolated ORT 1.23.2 renamed libmoonort.so; bundled deps: cpp-annotate, Eigen (MPL-2.0 subset), kaldi-native-fbank, KISS FFT, nlohmann/json, utf8, utf8proc.	MIT runtime and ORT; dependency licences (MIT, MPL-2.0, Apache-2.0, BSD-3-Clause, BSL-1.0, Unicode) in licenses/moonshine-runtime/. [1, 10]
Protocol Buffers javalite 4.36.1	Wire format for messages.	BSD-3-Clause. [11]
AndroidX, Jetpack Compose, Room, Kotlin, coroutines	App UI, storage and language runtime.	Apache-2.0; notices from resolved artifacts.
Android platform APIs	NSD, sockets, TLS/Conscrypt, AndroidKeyStore, BLE, Wi-Fi Direct.	OS-provided; nothing extra bundled.

Legacy recovery only: an English Parakeet 110M pack imported by an earlier build stays loadable so existing phones are not broken (CC BY 4.0 weights; licenses/parakeet-CC-BY-4.0.txt). It is not bundled or offered as a choice. Zipformer and Moonshine Base English packs are retired and the app refuses to activate them.

4. What is inside each app

CURRENT DEBUG APK	FILE SIZE	BUNDLED SPEECH COMPONENTS
Base	67.27 MB	eSpeak NG (all ten languages), Silero VAD and the runtimes. No STT models and no neural voices; import packs separately.
Preloaded	690.42 MB	Everything in base, plus English Small, English Tiny and Hindi STT, and the English and Hindi neural voices.

5. Licence foundations

All four are open-source licences approved by the Open Source Initiative. [13-16]

MIT: permits reuse with the copyright and licence notices retained. [14]

BSD-3-Clause: permits reuse with notices retained and without implying author endorsement. [16]

Apache-2.0: permits reuse with the licence, required notices and records of changed files. [15]

GPLv3: permits reuse while preserving source access and recipients' rights when distributing covered software. This is called **copyleft**. [13]

6. Why the whole app is GPL-3.0-or-later

iTantra links eSpeak NG as a native library called through JNI. The Free Software Foundation treats this as one combined program, so distribution conditions apply to the whole app even though the eSpeak library itself is unchanged. The project LICENSE is therefore GPL-3.0-or-later. MIT, BSD and Apache-2.0 components are compatible and keep their own notices. [12, 13]

Source access. Each distributed APK must be accompanied by the complete corresponding source: app and native code, the exact eSpeak NG source and voice-data inputs, the neural-voice export scripts, required dependency source and build instructions, at no extra charge (or included in USB handoffs). A repository URL alone does not discharge this. [13]

Attribution. THIRD_PARTY_NOTICES.md and the licenses/ folder ship in the APK assets and in each model pack, with the full publisher licence texts, credits and modification records. No endorsement by any upstream author is implied. [13-16]

7. Release documentation status

DOCUMENTATION	CURRENT STATUS
Recorded in the project	Pinned model versions, sizes, SHA-256 hashes, declared licences and third-party credits (models/models.lock.json, THIRD_PARTY_NOTICES.md, licenses/).
Pending release verification	Complete corresponding-source package; exact eSpeak NG acquisition revision; upstream confirmation of the AI4Bharat Indic-TTS MIT term; full resolved AndroidX/Kotlin notice inventory.

Licence basis: existing licence terms apply as published; no separate exception is assumed. This overview records the release plan, not a completed compliance audit.

Sources - click a reference to open

- [1] Moonshine: pinned model and code licence
- [2] AI4Bharat: IndicConformer and MIT licence
- [3] IndicConformer ONNX exports: model card
- [4] AI4Bharat: original Odia model declaration
- [5] AI4Bharat Indic-TTS: upstream repository
- [6] EchoBharat: int8 FastPitch conversions
- [7] eSpeak NG: upstream licence statement
- [8] Silero VAD: MIT licence

- [9] sherpa-onnx: Apache-2.0 licence
- [10] ONNX Runtime: MIT licence
- [11] Protocol Buffers: BSD-3-Clause licence
- [12] GNU FAQ: linking through JNI
- [13] GPLv3: source, notices and distribution
- [14] MIT: full licence and OSI approval
- [15] Apache-2.0: full licence and OSI approval
- [16] BSD-3-Clause: full licence and OSI approval

Evidence: current debug APKs, models/models.lock.json, THIRD_PARTY_NOTICES.md, LICENSE, licenses/, docs/NEURAL_TTS.md and DECISIONS.md. This is general information, not legal advice or SIH approval. Full licence terms apply. Model licences alone do not prove rights to all training data.